

Can action reconstruction make an editor obey?

Token-aligned sequence-edit JEPA: LDAD coefficient screen

17 July 2026

Low prediction error is not enough

- ▶ The model sees a damaged solution and a literal insert, delete, or replace action.
- ▶ It should predict the state produced by that particular edit.
- ▶ An action-blind model can predict an average nearby state and still obtain low error.
- ▶ We therefore swap the action at evaluation; a causal predictor should become at least 5% worse.

Question: does reconstructing the action from the observed latent change force the forward model to use its action?

Latent Difference Action Decoding adds an auxiliary task

- ▶ LDAD means Latent Difference Action Decoding.
- ▶ A decoder receives only the online latent difference between consecutive observed states.
- ▶ It reconstructs the complete observed edit phrase, including operation, position, and content.
- ▶ Five matched cells test LDAD weights 1, 10, and 20 and two representation-regularization controls.
- ▶ All runs use seed 0, 2,000 curriculum trajectories, three epochs, zero dropout, and 256 held-out examples.

LDAD improves dynamics but misses the causal gate

Condition	One-step	Recursive	Shuf./match	Rank
No VICReg, LDAD 10	.078	.140	1.007	137.6
VICReg .02, LDAD 1	.106	.190	1.010	131.2
VICReg .02, LDAD 10	.078	.140	1.007	137.5
VICReg .02, LDAD 20	.076	.134	1.006	137.9
VICReg .1, LDAD 10	.089	.159	1.011	136.8

- ▶ Every model beats persistence and reconstructs about 82% of action tokens.
- ▶ No shuffled-action ratio approaches the required 1.05 threshold.
- ▶ LDAD 20 has the best errors but a peak pre-clipping gradient norm of 241.7.

What to notice and why it matters

- ▶ The auxiliary decoder can read actions from observed state changes while the forward predictor still underuses the supplied action.
- ▶ LDAD is provisionally useful as a dynamics regularizer, not as a causal-action solution.
- ▶ Mask-only action sensitivity disappears when that checkpoint is evaluated on mixed edits, exposing a distribution confound.
- ▶ Non-curriculum runs also repeated corruptions despite a configuration requesting fresh examples; this is now fixed and tested.
- ▶ Next: matched fixed/fresh data controls, common cross-regime evaluation, and one lower-learning-rate LDAD-20 control.

Gate: no hierarchy or planning claim until mixed-edit action sensitivity reaches 1.05.